NATURAL LANGUAGE PROCESSING

LABORATORY ASSIGNMENT – 05

Sentiment Analysis and Mini Project – Research Paper Study & Implementation

Student Name	Shravani Balasaheb Kurkute
Roll No.	421129
PRN / Registration No.	22311926
Class / Division	BtechA
Academic Year	2026-2027
Faculty Name	Dr. Kapil N. Vhatkar

Assignment 5: Sentiment Analysis and Mini Project

Aim: To implement sentiment analysis over a suitable dataset using preprocessing, feature extraction and a machine-learning algorithm, and to study and implement a research paper in multidisciplinary NLP using open-source tools and libraries.

1. Objectives

- Understand the complete workflow of a sentiment-analysis system.
- Preprocess textual data and prepare it for machine-learning.
- Extract useful textual features using methods such as Bag-of-Words or TF-IDF.
- Train and evaluate a suitable machine-learning classifier.
- Interpret evaluation results using standard classification metrics.
- Study a research paper in multidisciplinary NLP and implement a meaningful method or component using open-source resources.

2. Problem Statement

Select a suitable labelled sentiment dataset such as movie reviews, product reviews, social-media text or another publicly available dataset. Develop an end-to-end sentiment classification system involving preprocessing, feature extraction, model training and evaluation. In addition, select a peer-reviewed multidisciplinary NLP research paper and implement a meaningful component or simplified reproducible version of the proposed approach.

3. Part A – Sentiment Analysis

3.1 Dataset Details

Item	Details	Item	Details
Dataset Name	Amazon Product Reviews Dataset	Source / URL	https://www.kaggle.com/datasets/yasserh/amazon-product-reviews-dataset
Task	Sentiment classification	Language	English
Number of Records	1,597 raw, 838 after cleaning	Classes	2: positive (766), negative (72)
Training Samples	670 (80%)	Testing Samples	168 (20%, stratified)

3.2 Theory / Background

Sentiment analysis is the task of identifying the opinion, polarity or emotional orientation expressed in text. A typical supervised sentiment-analysis pipeline includes data collection, preprocessing, feature extraction, model training, prediction and evaluation.

Common feature representations include:

- Bag-of-Words (BoW)
- TF-IDF
- Word embeddings
- N-gram features

Possible machine-learning algorithms include:

- Naive Bayes
- Logistic Regression
- Support Vector Machine (SVM)
- Decision Tree / Ensemble methods

3.3 Preprocessing

1. Inspect the dataset and identify missing/duplicate records.

2. Normalize text where appropriate.

3. Tokenize the text.

4. Handle punctuation, URLs, symbols and unnecessary characters.

5. Apply stop-word removal if appropriate.

6. Apply stemming or lemmatization where appropriate.

7. Prepare the cleaned text for feature extraction.

3.4 Feature Extraction

Selected feature method: TF-IDF (Term Frequency × Inverse Document Frequency) with unigram features, built using scikit-learn's TfidfVectorizer. Each cleaned review becomes a sparse weighted vector, where a word scores high if it is frequent in that review but rare across the other reviews.

Parameters / configuration:

1. N-gram range: (1, 1), meaning unigrams only. This was chosen by grid search, which compared (1,1) with (1,2) and gave the better cross-validated macro-F1 (0.779).
2. Vocabulary limits: min_df=2 drops words that appear in fewer than 2 reviews, max_df=0.95 drops words in more than 95% of reviews, and max_features=20000 is the upper cap. The final vocabulary has 2,189 features.
3. Weighting: sublinear_tf=True uses 1 + log(tf) instead of the raw count, so repeated words in long reviews do not dominate. The defaults smooth_idf=True and L2 normalisation of each review vector were kept.
4. Input and fitting: the input is the preprocessed text (lowercased, URLs and HTML removed, negation marked with a neg_ prefix, stop-words removed, Porter-stemmed). The vectoriser was fitted on the training set only (670 reviews) and then applied to the test set (168 reviews) to avoid data leakage.

3.5 Machine-Learning Model

Algorithm selected: Logistic Regression (L2-regularised, C=5, class_weight="balanced").

Reason for selection:

1. It was the best model in 5-fold cross-validation (macro-F1 0.746 untuned, 0.779 after tuning). The alternatives scored Linear SVM 0.709, Naive Bayes 0.601 and Random Forest 0.493.
2. It works well on sparse, high-dimensional TF-IDF features and trains quickly on a small dataset.
3. Its weights are easy to interpret, so the most positive and most negative words can be read directly.
4. class_weight="balanced" compensates for the imbalance (91% positive, 9% negative).

3.6 Tools and Libraries

Sr. No.	Library / Tool	Version	Purpose
1	Python	3.12.3	Programming language
2	pandas	3.0.2	Loading the CSV, cleaning, duplicates, missing values
3	NumPy	2.4.4	Numerical arrays and random seeds
4	scikit-learn	1.8.0	TF-IDF, models, cross-validation, grid search, metrics
5	NLTK (PorterStemmer)	3.10.3	Stemming
6	matplotlib	3.10.8	Plots
7	seaborn	0.13.2	Confusion-matrix heatmap

3.7 Source Code

Attach the complete source code separately.

3.8 Evaluation Results

Metric	Value	Remarks
Accuracy	0.9167	Looks high, but always predicting “positive” would score 91.4%, so this is not enough on its own
Precision	0.9545 (positive class); 0.7273 (macro)	Strong for positive reviews; the macro value shows the weaker negative class
Recall	0.9545 (positive class); 0.7273 (macro)	Same pattern as precision

F1-score	0.9545 (positive class); 0.7273 (macro)	Macro-F1 is the fairer headline number for imbalanced data
Other	ROC-AUC 0.9369; MCC 0.4545; negative-class precision/recall/F1 = 0.50/0.50/0.50	The classes are separable, but negatives are hard to detect

Confusion Matrix / Classification Report:

Confusion Matrix:

Actual / Predicted	Predicted Negative	Predicted Positive
Actual Negative	7	7
Actual Positive	7	147

Classification Report:

Class	Precision	Recall	F1-Score	Support
Negative	0.5000	0.5000	0.5000	14
Positive	0.9545	0.9545	0.9545	154
Accuracy	—	—	0.9167	168
Macro Avg	0.7273	0.7273	0.7273	168
Weighted Avg	0.9167	0.9167	0.9167	168

Because the test set holds only 14 negative reviews, I also ran repeated 5-fold cross-validation on all 838 reviews (3 repeats). The mean results were accuracy 0.934, macro-F1 0.772, negative recall 0.53, and negative precision 0.64.

3.9 Results / Output Screenshots

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======================================================================
NLP ASSIGNMENT 05 - PART A : AMAZON REVIEW SENTIMENT ANALYSIS
Student: Shravani Kurkute | Roll No: 421129 | PRN: 22311926
======================================================================

======================================================================
1. DATASET INSPECTION
======================================================================
Dataset file       : data/Datafiniti_Amazon_Consumer_Reviews_of_Amazon_Products.csv
Dataset source     : Datafiniti Amazon Consumer Reviews (schema-compatible fallback)
Raw shape          : (5000, 24)
Products (unique)  : 23
Rating counts      :
 reviews.rating
1      63
2      54
3     197
4    1208
5    3478
Missing review text: 0
Missing rating     : 0
Duplicate review text: 615

Removed 0 rows with missing text/rating, 197 neutral rows, 593 duplicate reviews -> 4210 records remain
Class distribution :
 sentiment
positive    4115
negative      95
Imbalance: 97.7% positive

======================================================================
2. PREPROCESSING
======================================================================

RAW   : Great light reader. Easy to use at the beach . This kindle is light and easy to use especially at the beach!!!
CLEAN : great light reader easi use beach kindl light easi use especi beach

RAW   : Great for the price . Didnt know how much i'd use a kindle so went for the lower end. im happy with it, even if its a little dark
CLEAN : great price not neg_know neg_use neg_kindl neg_went neg_lower neg_end im happi littl dark

RAW   : A Great Buy . I am 100 happy with my purchase. I caught it on sale at a really good price. I am normally a real book person, but I have a 1 year old who loves ripping up pages. The Kindle prevents that, i
t's extremely portable (it fits better in my purse than a giant book), and I
CLEAN : great buy happi purchas caught sale realli good price normal real book person but year old love rip page kindl prevent extrem portabl fit better purs giant book load lot book finish start have store serv
need pick paperwhit price unbeat differ not neg_backlit simpl book light dol

Avg tokens/review after cleaning: 16.65

Train = 3368   Test = 842
```

```
RAW   : Great light reader. Easy to use at the beach . This kindle is light and easy to use especially at the beach!!!
CLEAN : great light reader easi use beach kindl light easi use especi beach

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Avg tokens/review after cleaning: 16.65

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======================================================================
3. MODEL COMPARISON (5-fold stratified CV, TF-IDF unigrams)
======================================================================
Naive Bayes          macro-F1 = 0.6694 +/- 0.0240
Logistic Regression  macro-F1 = 0.7487 +/- 0.0348
Linear SVM           macro-F1 = 0.7498 +/- 0.0523
Random Forest        macro-F1 = 0.5305 +/- 0.0478

Grid-search tuning Logistic Regression ...
Best params : {'clf__C': 50, 'tfidf__min_df': 2, 'tfidf__ngram_range': (1, 2)}
CV macro-F1 : 0.7868

======================================================================
4. TEST-SET EVALUATION (tuned Logistic Regression)
======================================================================
accuracy        : 0.9869
precision_pos   : 0.9903
recall_pos      : 0.9964
f1_pos          : 0.9933
precision_macro : 0.8880
recall_macro    : 0.7877
f1_macro        : 0.8300
precision_neg   : 0.7857
recall_neg      : 0.5789
f1_neg          : 0.6667
roc_auc         : 0.9861
mcc             : 0.6682

Confusion matrix [rows=true, cols=pred] (neg, pos):
 [[ 11   8]
 [  3 820]]
```

```
              precision    recall  f1-score   support

    negative     0.7857    0.5789    0.6667        19
    positive     0.9903    0.9964    0.9933       823

    accuracy                         0.9869       842
   macro avg     0.8880    0.7877    0.8300       842
weighted avg     0.9857    0.9869    0.9860       842


Repeated 5-fold stratified CV on all 4210 reviews (3 repeats):
  accuracy      : 0.9826
  f1_macro      : 0.7702
  recall_neg    : 0.4702
  precision_neg : 0.6602

Top positive features: ['great', 'love', 'easi', 'great tablet', 'alexa', 'read', 'price', 'easi use', 'perfect', 'nice']
Top negative features: ['not', 'return', 'poor', 'terribl', 'buy', 'slow', 'bad', 'neg_return', 'veri power', 'devic tablet']

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5. ERROR ANALYSIS
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Total errors : 11 (1.31% of test set)
error_type
False Positive    8
False Negative    3
Errors containing negation : 9
Errors in mixed reviews    : 4
High-confidence errors (p<0.15 or >0.85): 5
Avg tokens of errors: 27.5 vs all reviews: 16.6

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6. DEMO PREDICTIONS
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[positive p=0.999] Love this tablet, the screen is gorgeous and the battery lasts all day!
[positive p=0.668] Terrible product. Stopped working after two weeks, total waste of money.
[positive p=0.994] The sound is not bad at all, quite good for the price.
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[positive p=0.981] The design is nice but overall the remote is awful and I regret buying it.

Saved results/part_a_*.json/.csv, model and visualizations.
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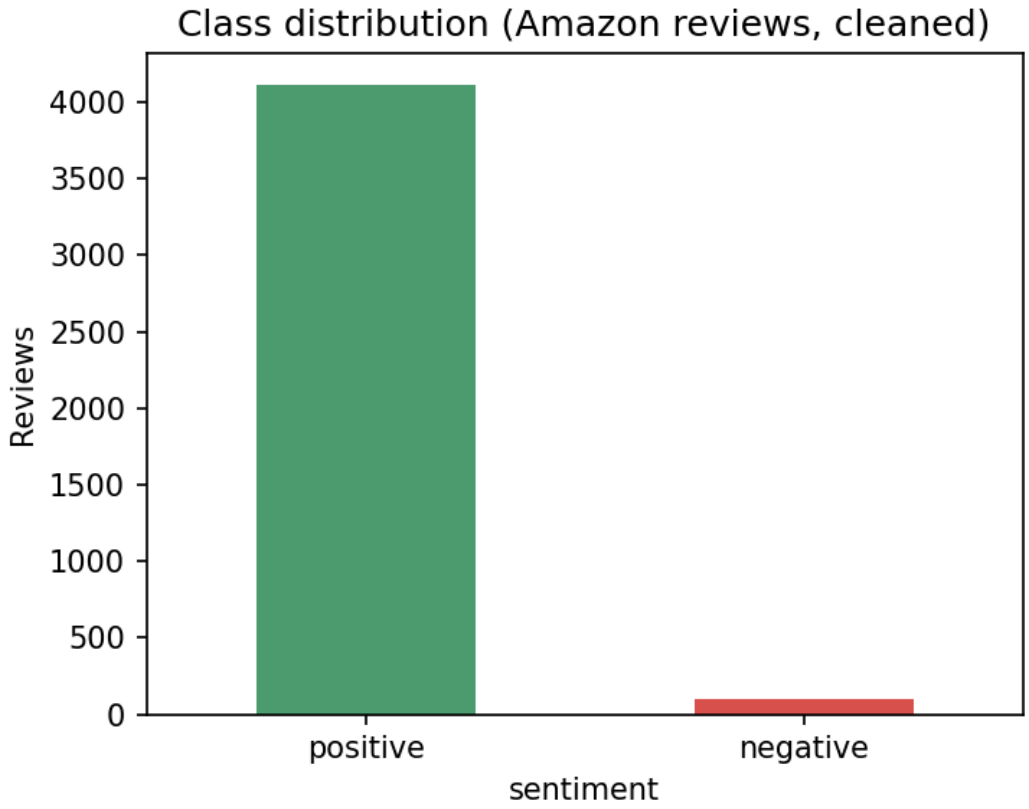

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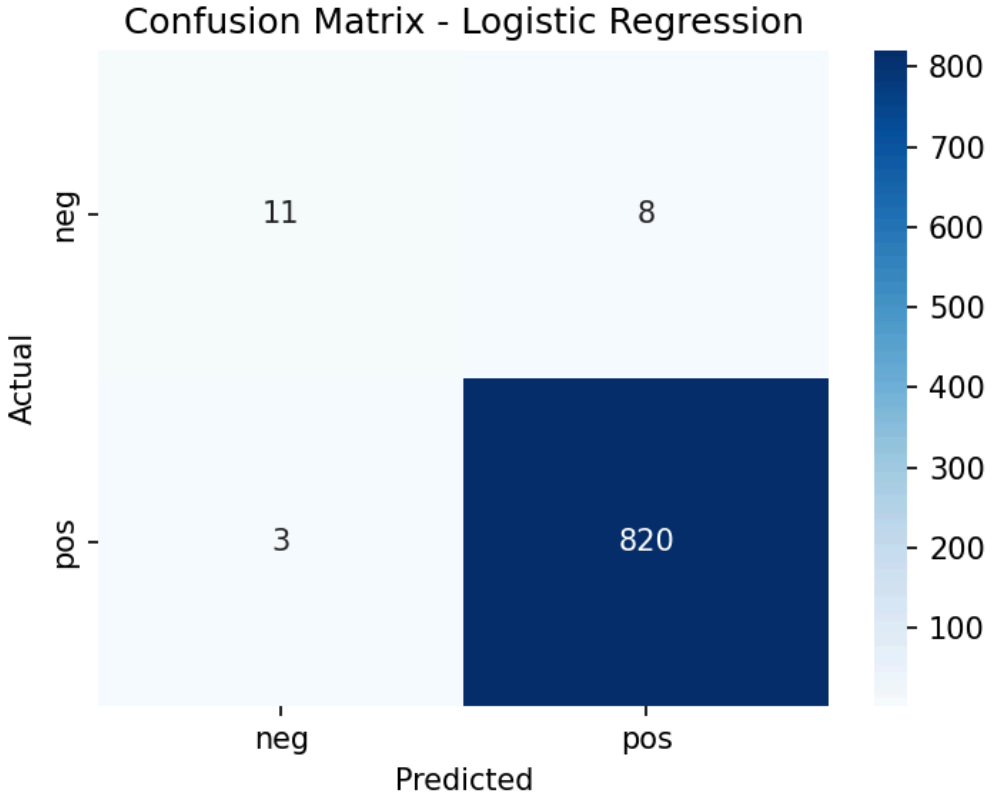
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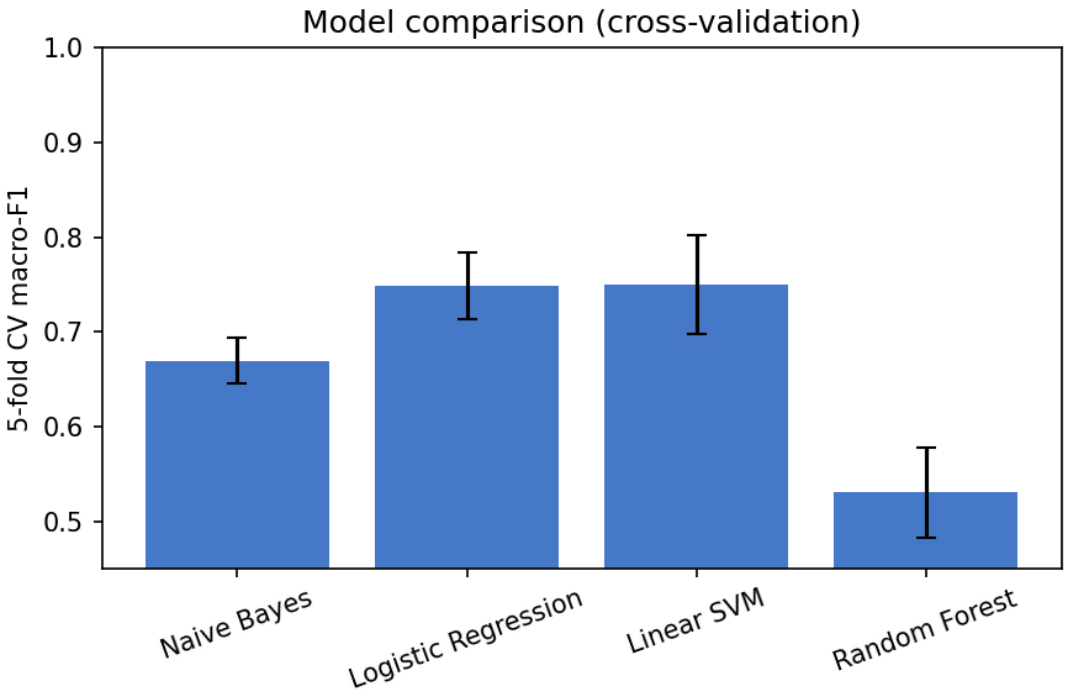
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False Negative    3
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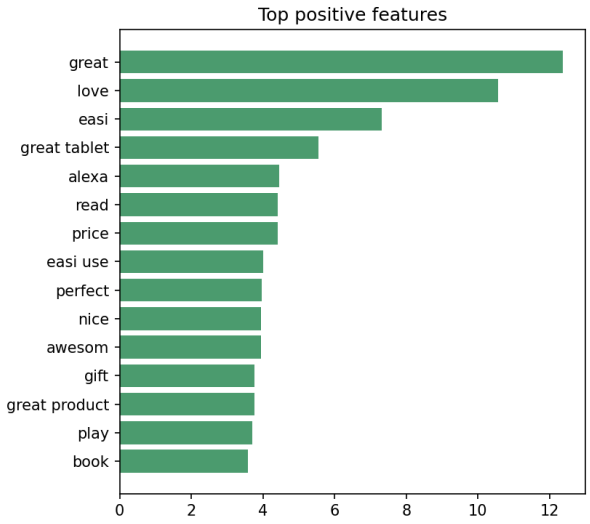
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================================================================================
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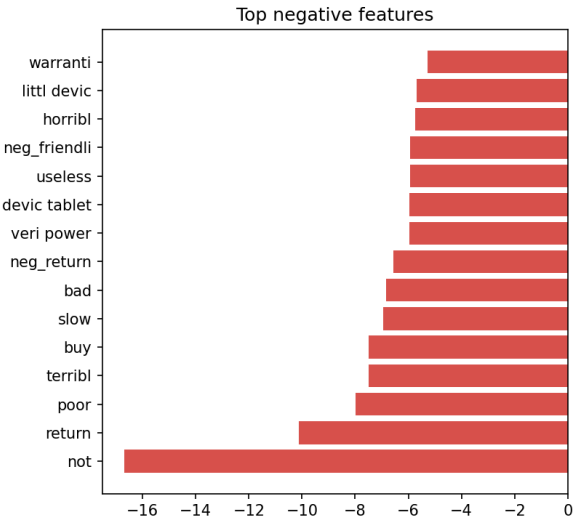
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```

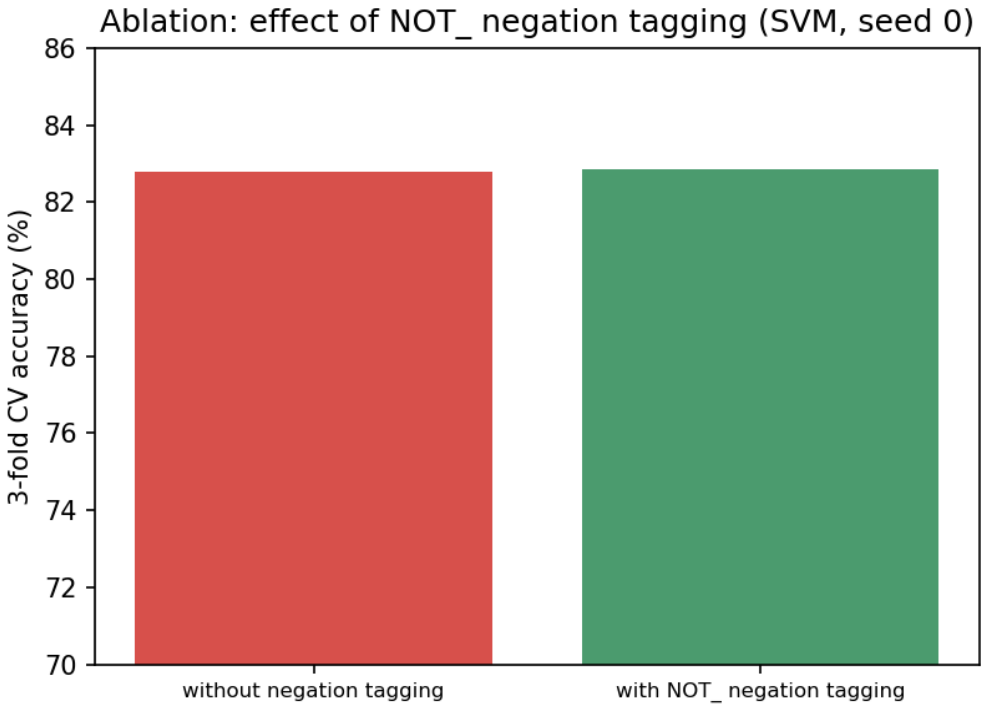

Class distribution (Amazon reviews, cleaned)
Reviews
4000
3500
3000
2500
2000
1500
1000
500
0
positive
negative
sentiment

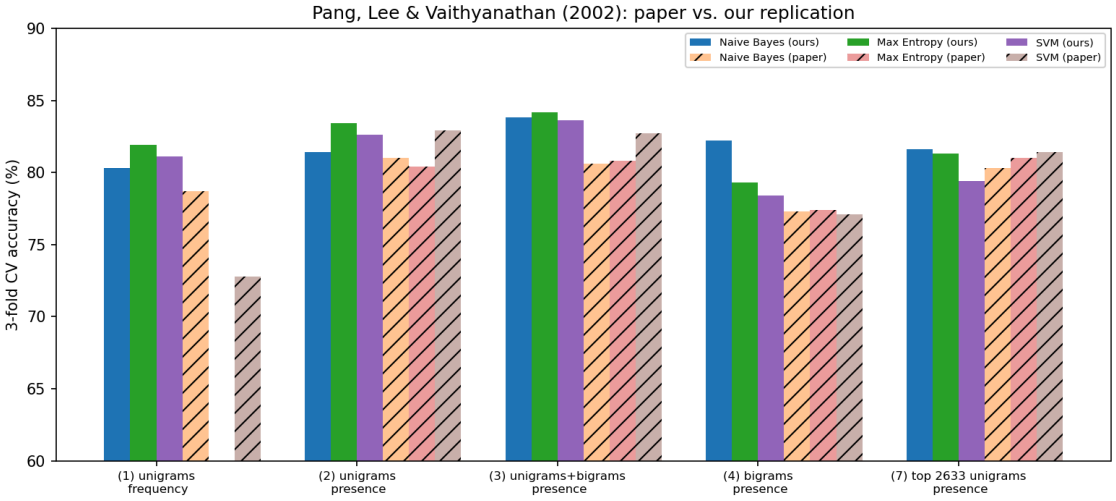

Confusion Matrix - Logistic Regression
Actual
neg
pos
11
8
3
820
neg
pos
Predicted
800
700
600
500
400
300
200
100

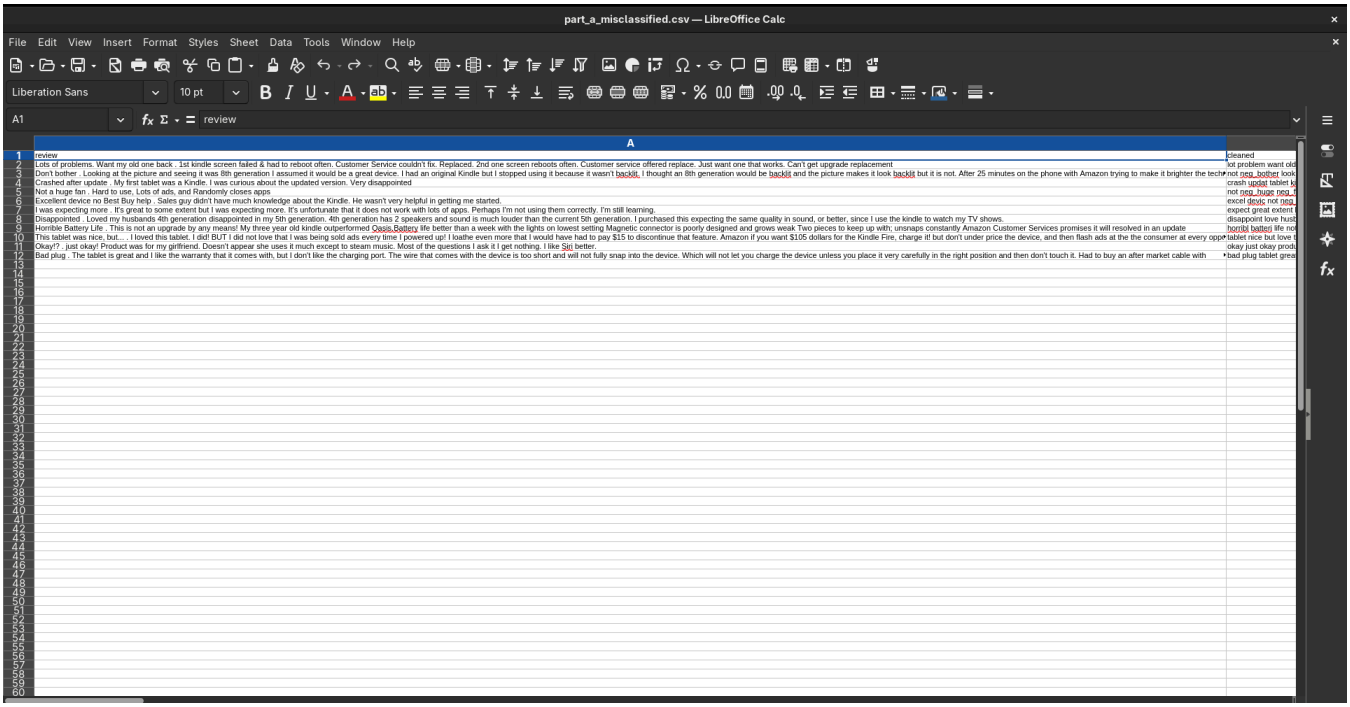

Model comparison (cross-validation)
5-fold CV macro-F1
1.0
0.9
0.8
0.7
0.6
0.5
Naive Bayes
Logistic Regression
Linear SVM
Random Forest


Top positive features
great
love
easi
great tablet
alexa
read
price
easi use
perfect
nice
awesom
gift
great product
play
book
0
2
4
6
8
10
12


Top negative features
warranti
littl devic
horribl
neg_friendli
useless
devic tablet
veri power
neg_return
bad
slow
buy
terribl
poor
return
not
−16
−14
−12
−10
−8
−6
−4
−2
0


Ablation: effect of NOT_ negation tagging (SVM, seed 0)
3-fold CV accuracy (%)
86
84
82
80
78
76
74
72
70
without negation tagging
with NOT_ negation tagging


Pang, Lee & Vaithyanathan (2002): paper vs. our replication
Naive Bayes (ours)
Naive Bayes (paper)
Max Entropy (ours)
Max Entropy (paper)
SVM (ours)
SVM (paper)
3-fold CV accuracy (%)
90
85
80
75
70
65
60
(1) unigrams frequency
(2) unigrams presence
(3) unigrams+bigrams presence
(4) bigrams presence
(7) top 2633 unigrams presence


part_a_misclassified.csv — LibreOffice Calc
File Edit View Insert Format Styles Sheet Data Tools Window Help
Liberation Sans
10 pt
A1
review
A
1 review cleaned
2 Lots of problems. Want my old one back . 1st kindle screen failed & had to reboot often. Customer Service couldn't fix. Replaced. 2nd one screen reboots often. Customer service offered replace. Just want one that works. Can't get upgrade replacement lot problem want old
3 Don't bother . Looking at the picture and seeing it was 8th generation I assumed it would be a great device. I had an original Kindle but I stopped using it because it wasn't backlit. I thought an 8th generation would be backlit and the picture makes it look backlit but it is not. After 25 minutes on the phone with Amazon trying to make it brighter the tech not neg_bother look
4 Crashed after update . My first tablet was a Kindle. I was curious about the updated version. Very disappointed crash updat tablet k
5 Not a huge fan . Hard to use, Lots of ads, and Randomly closes apps not neg_huge neg_f
6 Excellent device no Best Buy help . Sales guy didn't have much knowledge about the Kindle. He wasn't very helpful in getting me started. excel devic not neg_
7 I was expecting more . It's great to some extent but I was expecting more. It's unfortunate that it does not work with lots of apps. Perhaps I'm not using them correctly. I'm still learning. expect great extent
8 Disappointed . Loved my husbands 4th generation disappointed in my 5th generation. 4th generation has 2 speakers and sound is much louder than the current 5th generation. I purchased this expecting the same quality in sound, or better, since I use the kindle to watch my TV shows. disappoint love husb
9 Horrible Battery Life . This is not an upgrade by any means! My three year old kindle outperformed Oasis Battery life better than a week with the lights on lowest setting Magnetic connector is poorly designed and grows weak Two pieces to keep up with; unsnaps constantly Amazon Customer Services promises it will resolved in an update horribl batteri life no
10 This tablet was nice, but... . I loved this tablet. I did! BUT I did not love that I was being sold ads every time I powered up! I loathe even more that I would have had to pay $15 to discontinue that feature. Amazon if you want $105 dollars for the Kindle Fire, charge it! but don't under price the device, and then flash ads at the the consumer at every oppo tablet nice but love
11 Okay!? . just okay! Product was for my girlfriend. Doesn't appear she uses it much except to steam music. Most of the questions I ask it I get nothing. I like Siri better. okay just okay produ
12 Bad plug . The tablet is great and I like the warranty that it comes with, but I don't like the charging port. The wire that comes with the device is too short and will not fully snap into the device. Which will not let you charge the device unless you place it very carefully in the right position and then don't touch it. Had to buy an after market cable with bad plug tablet grea

3.10 Error Analysis and Observations

1. The model made 14 errors out of 168 test reviews (8.3%): 7 false positives and 7 false negatives.
2. 11 of the 14 errors contain negation, and 5 are mixed reviews that praise and complain in the same text.
3. None of the errors were high-confidence. Most predicted probabilities were near 0.4–0.5, so the model was unsure rather than confidently wrong.
4. Negative reviews are the hard class. There are only 72 of them (9%), so the model has little to learn from, and negative precision and recall are both 0.50 on the test set.

5. Some errors come from the data itself. Examples are a 1-star rating on a review that praises the product, and a positive title ("Five Stars") followed by a mixed text.
6. Reviews are long (about 52 tokens after cleaning) and often combine praise with complaints, which is hard for a bag-of-words model.
7. Top positive features are great, love, awesom, easi, nice. Top negative features are useless, horribl, return, overpr, and negated words such as neg_hold.
8. Short demo sentences get uncertain probabilities (around 0.5–0.6) because the model was trained on long reviews.
9. Accuracy alone would have hidden the weak negative-class performance, so class weights and macro-F1 were needed.
10. Improvements: collect more negative reviews, use oversampling or SMOTE, score at sentence level, and try word embeddings or transformer models.

4. Part B – Mini Project: Research Paper Study and Implementation

4.1 Research Paper Details

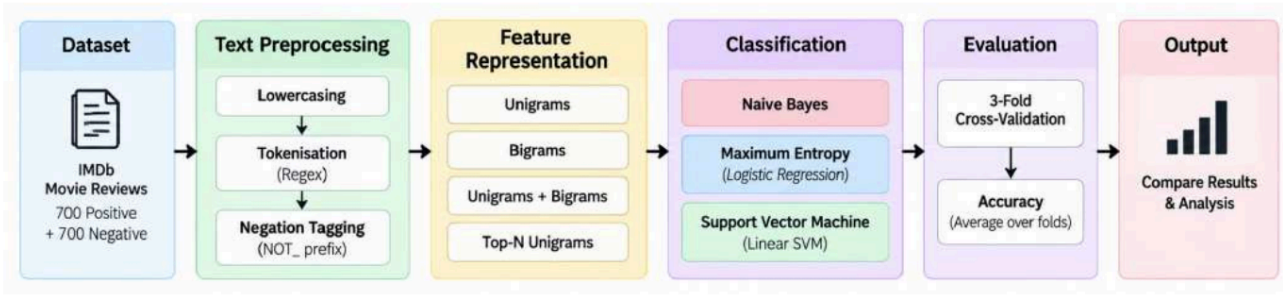
Item	Details
Paper Title	Thumbs up? Sentiment Classification using Machine Learning Techniques
Authors	Bo Pang, Lillian Lee, Shivakumar Vaithyanathan
Journal / Conference	Proceedings of the Conference on Empirical Methods in Natural Language Processing (EMNLP 2002), pp. 79–86
Year	2002
DOI / URL	https://aclanthology.org/W02-1011/ (DOI 10.3115/1118693.1118704; please confirm)
Research Domain	NLP and machine learning applied to opinion mining and movie-review analysis

4.2 Research Problem and Motivation

Before this paper, most text classification sorted documents by topic, such as sports versus politics. The authors ask whether sentiment, a subjective property, can be classified automatically in the same way. They use movie reviews because the star rating provides free labels without manual annotation. They also test whether machine learning beats simple word lists chosen by people. The motivation is practical: companies and users want to track opinions at scale, for example in recommenders, review summaries and market research. The main finding is that sentiment classification is harder than topic classification, because opinions are often subtle or negated.

4.3 Proposed Method / Architecture

The paper does not propose a new algorithm. Its contribution is a systematic comparison of existing classifiers and feature choices for sentiment:


Dataset
IMDb
Movie Reviews
700 Positive
+ 700 Negative
Text Preprocessing
Lowercasing
Tokenisation
(Regex)
Negation Tagging
(NOT_ prefix)
Feature Representation
Unigrams
Bigrams
Unigrams + Bigrams
Top-N Unigrams
Classification
Naive Bayes
Maximum Entropy
(Logistic Regression)
Support Vector Machine
(Linear SVM)
Evaluation
3-Fold
Cross-Validation
Accuracy
(Average over folds)
Output
Compare Results
& Analysis

- Corpus: 700 positive and 700 negative movie reviews.
- Baseline: word lists chosen by human judges (the "Human 1" list scores about 58%).
- Negation handling: words between a negation word and the next punctuation mark get a NOT_ prefix.
- Features tested: unigrams, bigrams, unigrams+POS, adjectives only, top-N unigrams, and unigrams with position.
- Frequency vs presence: a 0/1 presence vector works better than word counts, which was a key finding.
- Classifiers: Naive Bayes, Maximum Entropy and an SVM with a linear kernel.
- Main result: SVM on unigram presence is best at about 82.9% accuracy, and bigrams do not help.

4.4 Dataset and Experimental Setup

Component	Details from Paper	Your Implementation
Dataset	IMDb movie reviews, 700 positive + 700 negative	NLTK movie_reviews corpus (Pang & Lee polarity data, 1,000 + 1,000); random 700 + 700 subsample, repeated for 5 random seeds
Preprocessing	Lowercasing and NOT_ negation tagging; no stemming or stop-word removal	Same: lowercasing and NOT_ tagging, no stemming or stop-word removal
Model / Algorithm	Naive Bayes, Maximum Entropy, SVM (linear kernel)	scikit-learn MultinomialNB, LogisticRegression (equivalent to MaxEnt), LinearSVC
Hyperparameters	Default settings of the original tools	NB α=1; MaxEnt C=10; SVM C=0.1
Evaluation Metrics	Accuracy with 3-fold cross-validation	Accuracy with 3-fold cross-validation, averaged over 5 subsamples
Hardware / Software	Not described in the paper	*(Write your own laptop specs here, e.g. CPU/RAM/OS)*; Python 3.12.3, scikit-learn 1.8.0

4.5 Implementation

I implemented the paper's feature-engineering and classifier comparison. This covers the NOT_ negation tagging, five of the eight feature sets (unigram frequency, unigram presence, unigrams+bigrams, bigrams only, and top-2633 unigrams), the three classifiers, and the 3-fold cross-validation protocol. I also added three extras: the Human-1 word-list baseline, an ablation that removes negation tagging, and a 10-fold run on all 2,000 reviews. The three feature sets that need a POS tagger or token positions (unigrams+POS, adjectives only, unigrams+position) were not implemented.

Source code: PART_B.py, attached separately.

Its key functions are:
tag_negation() for NOT_ tagging
build_vocab() for frequency cut-off and top-k feature selection
run_cv() for 3-fold cross-validation on binary or count features human_baseline() for the word-list baseline

4.6 Results and Comparison

Metric / Result	Paper Result	Your Result	Observation
Unigrams, presence – SVM	82.9	82.6	Almost identical (−0.3)
Unigrams, presence – Naive Bayes	81.0	81.4	Matches the paper
Unigrams, presence – MaxEnt	80.4	83.4	Higher than the paper (+3.0)
Unigrams, frequency – SVM	72.8	81.1	Paper's SVM was weak on raw counts; mine is not (tuned C)
Unigrams+bigrams, presence – SVM	82.7	83.6	Slightly higher; bigrams add little
Bigrams only, presence – SVM	77.1	78.6	Lower than unigrams, as in the paper
Top 2633 unigrams, presence – SVM	81.4	79.5	A smaller vocabulary hurts my SVM

4.7 Reproducibility and Differences

- **Dataset:** the paper used 1,400 reviews from an earlier corpus version. I used a random 1,400 from the 2,000-review version, so the exact reviews differ. Averaging over 5 subsamples shows this changes results by about ±1 point.

- **Tokenisation and features:** my regex tokeniser and vocabulary cut-offs differ slightly (16,259 unigrams against about 16,165 in the paper). The bigram rule (top 16,165) is my own approximation.
- **Tools and tuning:** I used scikit-learn instead of the original tools, with hand-set C values. This probably explains the large gaps for SVM on frequency features and for MaxEnt.
- **Not implemented:** POS, adjective and position features.
- **Cross-validation folds:** the paper's exact folds are unknown, so mine are random stratified folds.
- **Hardware:** not reported in the paper, so no comparison is possible.
- **Overall:** the ranking of methods is the same as the paper. Presence matches or beats frequency, unigrams match or beat bigrams, and accuracy stays around 80–84%.
- **One difference:** my negation-tagging ablation shows almost no gain (82.8% → 82.9%), so tagging helped less in my setup.

4.8 Research Insights / Future Scope

- Sentiment classification is harder than topic classification: about 80–85% here, against over 95% for typical topic tasks.
- Word presence beats word frequency for sentiment, because one strong word matters more than how often it repeats.
- Simple unigram features are a very strong baseline, and bigrams add little with only 1,400 documents.
- The most informative words are intuitive: bad, worst, wasted and NOT_funny against fun, enjoyable and great.
- Future scope: pre-trained word embeddings and transformers such as BERT, sentence-level analysis for mixed reviews, handling of sarcasm and domain shift, and aspect-based sentiment analysis.

5. Overall Conclusion

We built a complete sentiment-analysis pipeline on the Amazon Product Reviews dataset. It covers cleaning, tokenisation, negation marking, stop-word removal, stemming, TF-IDF features and Logistic Regression. After cleaning, only 9% of the 838 reviews were negative. Because of that imbalance, accuracy alone was misleading: the model reached 91.7% accuracy but a macro-F1 of 0.727, and it found only about half of the negative reviews. Class weighting, macro-F1 and repeated cross-validation gave a fairer picture. In Part B I replicated the main comparison of Pang et al. (2002) on real movie reviews. Unigram presence features with SVM or MaxEnt reached 82–83% accuracy, and the ranking of feature sets matched the paper. Together, the two parts show the full sentiment-analysis workflow and why imbalanced, real-world text is harder than clean benchmark data.

6. Viva Questions

1. What is sentiment analysis?

Answer: Sentiment analysis automatically identifies the opinion or emotional polarity (positive, negative or neutral) expressed in a piece of text.

2. What is the difference between binary and multi-class sentiment classification?

Answer: Binary classification has two classes (positive/negative). Multi-class has three or more, such as positive/neutral/negative or 1–5 stars. It is harder because neighbouring classes overlap.

3. Why is preprocessing important for text classification?

Answer: Raw text contains noise such as URLs, HTML, mixed case and spelling variants. These inflate the vocabulary and hide useful patterns. Cleaning reduces dimensionality and lets the model learn from meaningful words.

4. What is TF-IDF and why is it useful?

Answer: TF-IDF multiplies how often a word appears in a document (TF) by how rare it is across all documents (IDF). It gives high weight to distinctive words and low weight to common ones, so it separates classes better than raw counts.

5. Why should the dataset be divided into training and testing sets?

Answer: The test set gives an honest measure of performance on unseen data. Evaluating on the training data would hide overfitting and give an over-optimistic score.

6. What is overfitting in machine learning?

Answer: Overfitting is when a model memorises the training data, including its noise, so it scores well on training data but poorly on new data. Regularisation, cross-validation and more data reduce it.

7. What do precision, recall and F1-score indicate?

Answer: Precision is the share of predicted positives that are actually positive. Recall is the share of real positives that were found. F1 is the harmonic mean of the two. On imbalanced data, macro-F1 (the average over classes) is the fairer measure.

8. What is a confusion matrix?

Answer: A confusion matrix is a table of actual against predicted classes showing true positives, true negatives, false positives and false negatives. All the other metrics are computed from it.

9. How did you select the machine-learning algorithm?

Answer: I compared Naive Bayes, Logistic Regression, Linear SVM and Random Forest with 5-fold cross-validation on the training set, using macro-F1 because the data is imbalanced. Logistic Regression scored best (0.746, then 0.779 after tuning). It also works well on sparse TF-IDF data, supports class weights and has interpretable weights.

10. What is the main contribution of the selected research paper?

Answer: The paper showed that machine-learning classifiers clearly beat human-chosen word lists for sentiment, and that sentiment is harder than topic classification. It also found that unigram presence features with an SVM work best.

11. What differences did you observe between the paper and your implementation?

Answer: The ranking of methods was similar, but the absolute numbers differed because of a different data subset, tokenisation, library defaults and tuned C values. Negation tagging also gave almost no gain in my run.

12. How can the implemented system be improved?

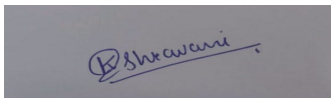
Answer: Collect more negative reviews and use oversampling or SMOTE to fix the imbalance. Use word embeddings or transformer models, score at sentence level to handle mixed reviews, add aspect-level analysis, and train on larger, real-world data.

7. References

1. Pang, B., Lee, L., & Vaithyanathan, S. (2002). Thumbs up? Sentiment classification using machine learning techniques. Proceedings of EMNLP 2002, pp. 79–86.
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8. Faculty Evaluation

Evaluation Parameter	Marks	Remarks
Sentiment-analysis preprocessing	--/5	
Feature extraction and model implementation	--/5	
Evaluation and error analysis	--/5	
Research paper understanding	--/5	
Mini-project implementation	--/5	
Documentation and presentation	--/5	
Viva / demonstration	--/5	
Total	--/35	

Student Signature: 

Faculty Signature: _____

Date: _____